Fermi's golden rule

Find the transition rate $\Gamma_{1\to 2}$ for the perturbing Hamiltonian

$$H_1(x,t) = 2V(x)\cos(\omega t + \phi)$$

Let $\Psi(x,t)$ be the following linear combination of two eigenstates.

$$\Psi(x,t) = c_1(t)\psi_1(x)\exp(-i\omega_1 t) + c_2(t)\psi_2(x)\exp(-i\omega_2 t)$$

We need to solve for $c_2(t)$ to find the transition rate $\Gamma_{1\to 2}$.

From the time-dependent Schrodinger equation we have

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = H_0(x)\Psi(x,t) + H_1(x,t)\Psi(x,t)$$

Expand the left-hand side.

$$\begin{aligned} i\hbar\frac{\partial}{\partial t}\Psi(x,t) = i\hbar\frac{\partial c_1(t)}{\partial t}\psi_1(x)\exp(-i\omega_1 t) &+ \boxed{\hbar\omega_1 c_1(t)\psi_1(x)\exp(-i\omega_1 t)} \\ + i\hbar\frac{\partial c_2(t)}{\partial t}\psi_2(x)\exp(-i\omega_2 t) &+ \boxed{\hbar\omega_2 c_2(t)\psi_2(x)\exp(-i\omega_2 t)} \end{aligned}$$

Noting that $\hbar\omega_1 = E_1$ and $\hbar\omega_2 = E_2$, the boxed terms cancel with $H_0(x)\Psi(x,t)$ leaving

$$i\hbar\frac{\partial c_1(t)}{\partial t}\psi_1(x)\exp(-i\omega_1 t) + i\hbar\frac{\partial c_2(t)}{\partial t}\psi_2(x)\exp(-i\omega_2 t) = H_1(x,t)\Psi(x,t)$$

Substitute for $H_1(x)\Psi(x,t)$.

$$\begin{aligned} &i\hbar\frac{\partial c_1(t)}{\partial t}\psi_1(x)\exp(-i\omega_1 t) + i\hbar\frac{\partial c_2(t)}{\partial t}\psi_2(x)\exp(-i\omega_2 t) \\ &\qquad = 2\cos(\omega t + \phi)\Big[c_1(t)V(x)\psi_1(x)\exp(-i\omega_1 t) + c_2(t)V(x)\psi_2(x)\exp(-i\omega_2 t)\Big] \end{aligned}$$

Take the inner product of $\psi_2^*(x)$ with both sides to obtain

$$i\hbar\frac{\partial c_2(t)}{\partial t}\exp(-i\omega_2 t) = 2\cos(\omega t + \phi)\Big[c_1(t)M_{21}\exp(-i\omega_1 t) + c_2(t)M_{22}\exp(-i\omega_2 t)\Big]$$

Symbols M_{21} and M_{22} are the following matrix elements.

$$M_{21} = \int \psi_2^*(x)V(x)\psi_1(x)\,dx$$

$$M_{22} = \int \psi_2^*(x)V(x)\psi_2(x)\,dx$$

Multiply both sides by $\exp(i\omega_2 t)$.

$$i\hbar\frac{\partial c_2(t)}{\partial t} = 2\cos(\omega t + \phi)\Big[c_1(t)M_{21}\exp[i(\omega_2 - \omega_1)t] + c_2(t)M_{22}\Big]$$

Let the initial state be $\Psi(x,0) = \psi_1(x)$ hence the initial conditions are

$$c_1(0) = 1, \quad c_2(0) = 0$$

For time t near the origin use the approximations $c_1(t) = 1$ and $c_2(t) = 0$ to obtain

$$i\hbar \frac{\partial c_2(t)}{\partial t} = 2\cos(\omega t + \phi) M_{21} \exp[i(\omega_2 - \omega_1)t]$$

Solve for $c_2(t)$ by integrating.

$$c_2(t) = \frac{2M_{21}}{i\hbar} \int_0^t \cos(\omega t' + \phi) \exp[i(\omega_2 - \omega_1)t']\, dt'$$

In exponential form

$$\begin{aligned} c_2(t) = \frac{M_{21}}{i\hbar} \int_0^t \exp(i\omega t' + \phi) \exp[i(\omega_2 - \omega_1)t']\, dt' \\ + \frac{M_{21}}{i\hbar} \int_0^t \exp(-i\omega t' - \phi) \exp[i(\omega_2 - \omega_1)t']\, dt' \end{aligned}$$

The solution is

$$\begin{aligned} c_2(t) = -\frac{M_{21}}{\hbar} \frac{\exp[i(\omega_2 - \omega_1 + \omega)t] - 1}{\omega_2 - \omega_1 + \omega} \exp(i\phi) \\ - \frac{M_{21}}{\hbar} \frac{\exp[i(\omega_2 - \omega_1 - \omega)t] - 1}{\omega_2 - \omega_1 - \omega} \exp(-i\phi) \end{aligned} \tag{1}$$

For ω such that $\omega \approx \omega_2 - \omega_1$ the second term dominates so discard the first term.

$$c_2(t) = -\frac{M_{21}}{\hbar} \frac{\exp[i(\omega_2 - \omega_1 - \omega)t] - 1}{\omega_2 - \omega_1 - \omega} \exp(-i\phi)$$

By the identity

$$\exp(ix) - 1 = 2i \sin\left(\frac{x}{2}\right) \exp\left(\frac{ix}{2}\right)$$

we have

$$c_2(t) = -\frac{2iM_{21}}{\hbar} \frac{\sin\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right] \exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right]}{\omega_2 - \omega_1 - \omega} \exp(-i\phi)$$

Multiply numerator and denominator by $t/2$.

$$c_2(t) = -\frac{itM_{21}}{\hbar} \frac{\sin\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right] \exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right]}{\frac{1}{2}(\omega_2 - \omega_1 - \omega)t} \exp(-i\phi)$$

Substitute $\operatorname{sinc}(x)$ for $\sin(x)/x$.

$$c_2(t) = -\frac{itM_{21}}{\hbar} \operatorname{sinc}\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right] \exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right] \exp(-i\phi) \tag{2}$$

Hence the transition probability is

$$P(1 \to 2) = |c_2(t)|^2 = \frac{t^2}{\hbar^2} |M_{21}|^2 \operatorname{sinc}^2 \left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t \right] \tag{3}$$

Integrate $P(1 \to 2)$ to obtain the total transition probability $P_{tot}(1 \to 2)$.

$$P_{tot}(1 \to 2) = \frac{t^2}{\hbar^2} |M_{21}|^2 \int_{E-\epsilon}^{E+\epsilon} \operatorname{sinc}^2 \left(\frac{E'/\hbar - \omega}{2} t \right) g(E') \, dE'$$

where

$$E = \hbar(\omega_2 - \omega_1)$$

and $g(E')$ is the density of photon states for energy E'.

Use the approximation $g(E') \approx g(\hbar\omega)$ to obtain

$$P_{tot}(1 \to 2) = \frac{t^2}{\hbar^2} |M_{21}|^2 g(\hbar\omega) \int_{E-\epsilon}^{E+\epsilon} \operatorname{sinc}^2 \left(\frac{E'/\hbar - \omega}{2} t \right) dE'$$

Let

$$y = \frac{E'/\hbar - \omega}{2} t$$

It follows that

$$E' = \frac{2\hbar y}{t} + \hbar\omega$$

and

$$dE' = \frac{2\hbar}{t} \, dy$$

The integration limits transform as

$$E \pm \epsilon \to \frac{(E \pm \epsilon)/\hbar - \omega}{2} t = \frac{Et}{2\hbar} - \frac{\omega t}{2} \pm \frac{\epsilon t}{2\hbar} \approx \pm \frac{\epsilon t}{2\hbar}$$

Hence

$$P_{tot}(1 \to 2) = \frac{2t}{\hbar} |M_{21}|^2 g(\hbar\omega) \int_{-\epsilon t/2\hbar}^{\epsilon t/2\hbar} \operatorname{sinc}^2 y \, dy$$

Use the approximation

$$\int_{-\epsilon t/2\hbar}^{\epsilon t/2\hbar} \operatorname{sinc}^2 y \, dy \approx \int_{-\infty}^{\infty} \operatorname{sinc}^2 y \, dy = \pi$$

to obtain

$$P_{tot}(1 \to 2) = \frac{2\pi t}{\hbar} |M_{21}|^2 g(\hbar\omega)$$

The transition rate is the derivative of $P_{tot}(1 \to 2)$.

$$\Gamma_{1 \to 2} = \frac{d}{dt} P_{tot}(1 \to 2) = \frac{2\pi}{\hbar} |M_{21}|^2 g(\hbar\omega)$$